VOICE-NAVIGATION

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING BY

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PRASAD V POTLURI SIDDHARTHA INSTITUTE OF TECHNOLOGY

(Permanently affiliated to JNTU :: Kakinada, Approved by AICTE)

(An NBA & NAAC A+ accredited and ISO 9001:2015 Certified Institution)

Kanuru, Vijayawada - 520007

(2024-25)

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CERTIFICATE

This is to certify that the project report titled **"VOICE-NAVIGATION"** is the bona fide work of **V. Phani Sirisha (22501A05J2), V. Sumanth (22501A05I8), P. Mahesh (22501A05E2), P. Sunanda (22501A05F1)** in partial fulfilment of completing the Academic project in Mobile App Development Lab during the academic year 2024-25.

Signature of the Incharge **Signature of the HOD**

INDEX

1.1 SDG JUSTIFICATION REPORT

SDG Mapped: SDG 3 – Good Health and Well-being, SDG 4 – Quality Education

1.1.1 How the Accessible India Supports SDGs 3 & 4

The **Accessible India (Sugamya Bharat Abhiyan)** directly contributes to achieving **SDG 3** and **SDG 4** by fostering inclusivity, accessibility, and equitable opportunities for persons with disabilities (PwDs) through:

- **Improved Access to Healthcare (SDG 3):**
  - Ensures barrier-free access to healthcare facilities, enabling PwDs to receive timely and appropriate medical services.
  - Promotes mental and physical well-being by fostering an inclusive environment.
  - Encourages the use of assistive technologies and accessible medical information.
- **Inclusive Quality Education (SDG 4):**
  - Ensures accessibility in educational institutions through ramps, braille materials, and assistive devices.
  - Promotes inclusive digital education platforms catering to diverse needs.
  - Offers vocational training and skill development opportunities, empowering PwDs for lifelong learning.
- **Creating a Barrier-Free Environment:**
  - Focuses on making public spaces, transportation, and digital platforms accessible to all, enabling PwDs to participate fully in society.
  - Drives awareness and policy changes for a more inclusive and accessible India.
- **Scalability for Broader Impact:**
  - The campaign's framework can be adopted across cities, creating an inclusive ecosystem that supports the well-being, education, and empowerment of PwDs on a national scale.

The **Accessible India** embodies the principle of "Leave No One Behind" by removing barriers and ensuring equal opportunities for all.

1. ABSTRACT

In today’s fast-paced world, independence and mobility are essential for leading a confident and fulfilling life. However, visually impaired individuals often face challenges in navigating unfamiliar environments. **Voice-Navigation** aims to bridge this gap by offering an intuitive and user-friendly mobile application that empowers visually impaired individuals to travel independently and safely.

The inspiration behind **Voice-Navigation: Empowering Mobility for All** stems from the need to create an inclusive platform that provides real-time navigation through voice-guided assistance. The application supports two types of users:

- **Travelers:** Can initiate voice-based navigation, choose starting points through tapping, and receive step-by-step audio instructions for seamless mobility.

Key Features of Voice-Navigation:

- **Voice-to-Text Input:** Allows users to input source and destination through voice commands, ensuring ease of use.

- **Tap-to-Select Start Point:** Users can manually select their starting point by tapping anywhere on the screen.

- **Audio Navigation Cues:** Provides step-by-step voice instructions to guide users safely to their destination. `

Beyond being a navigation tool, **Voice-Navigation** promotes independence and confidence by offering a secure and reliable platform that empowers visually impaired individuals to explore the world with ease.

By incorporating innovative features and focusing on user-centric design, **Voice-Navigation** not only enhances mobility but also fosters inclusivity and empowerment. The application leverages cutting-edge technology to provide a seamless navigation experience, ensuring that visually impaired users can navigate both familiar and unfamiliar environments comfortably and confidently.

2. INTRODUCTION

In today's technology-driven world, mobility and independence are essential for ensuring equal opportunities and quality of life for visually impaired individuals. However, navigating unfamiliar environments can be a daunting task, often limiting their ability to explore and engage with the world around them. **Voice-Navigation** is a mobile application designed to address this challenge by providing an intuitive and accessible platform that offers real-time, voice-guided navigation to help visually impaired individuals travel independently and safely.

Voice-Navigation leverages advanced technologies to offer a seamless and user-friendly experience. Users can initiate navigation by using **voice-to-text input** to specify their source and destination, ensuring ease of use without requiring manual typing. Additionally, the **tap-to-select start point** feature allows users to manually choose their starting location by tapping anywhere on the screen, offering flexibility when needed. Once the destination is set, the application provides **audio navigation cues** with step-by-step voice instructions, guiding users safely and accurately to their desired destination.

Beyond its core functionality, **Voice-Navigation** is designed with a focus on accessibility and ease of use. The interface is simple, intuitive, and optimized to ensure smooth interaction for visually impaired users. The combination of voice commands and touch-based input makes the application versatile, catering to different user preferences and scenarios. Furthermore, the audio navigation cues are carefully designed to provide clear, concise, and timely instructions, ensuring that users remain aware of their surroundings and navigate safely. By integrating these features, **Voice-Navigation** not only enhances mobility but also promotes independence, confidence, and inclusivity, enabling visually impaired individuals to explore the world with greater ease.

3. OBJECTIVES AND SCOPE OF THE PROJECT

Objectives

The primary objectives of **Voice-Navigation** are:

1. **Empower Visually Impaired Individuals:** Provide an intuitive and reliable navigation solution that enables visually impaired users to travel independently and confidently.
2. **Seamless Voice Input:** Implement a **voice-to-text** system that allows users to input their source and destination effortlessly through voice commands.
3. **Flexible Start Point Selection:** Incorporate a **tap-to-select start point** feature, allowing users to manually choose their starting location if needed.
4. **Real-Time Audio Guidance:** Deliver **audio navigation cues** that provide step-by-step instructions to guide users safely to their destination.
5. **Promote Safety and Inclusivity:** Ensure that visually impaired individuals can navigate unfamiliar environments safely while fostering inclusivity through accessible technology.

Scope of the Project

The scope of **Voice-Navigation** includes:

1. **User Interface Design:** Develop a clean, user-friendly interface optimized for visually impaired users, ensuring smooth interaction with voice commands and touch-based input.
2. **Voice-to-Text System:** Implement accurate and responsive voice recognition technology to process user commands and initiate navigation efficiently.
3. **Tap-to-Select Start Point Module:** Enable users to manually set their starting point by tapping on the screen, providing flexibility in various situations.
4. **Audio Navigation System:** Integrate a reliable audio guidance system that delivers clear and concise directions throughout the journey.
5. **Route Optimization:** Ensure efficient route calculation and guidance, minimizing travel time and avoiding obstacles.
6. **Cross-Platform Compatibility:** Develop the application to be compatible with both Android and iOS devices, ensuring accessibility to a wide range of users.
7. **Scalability and Upgradability:** Design the application architecture to allow future enhancements, including additional features and improved navigation capabilities.

4. SOFTWARE USED

The **Voice-Navigation** application is developed using a combination of technologies and tools that ensure efficient performance, accuracy, and a seamless user experience. The key software and tools used in the project include:

1. Programming Language

- **Java:** The core language used for Android application development, ensuring robust and efficient implementation.

2. Development Platform

- **Android Studio:** The official IDE for Android development, used to design, develop, and test the application.

3. API and Libraries

- **Google Maps API:** Enables location-based services and facilitates real-time navigation with accurate route guidance.
- **SpeechRecognizer API:** Provides the speech-to-text functionality that converts user voice input into text for processing source and destination.
- **RecognizerIntent Class:** Used to initiate speech recognition and receive results from the system.

4. User Interface Design

- **XML:** Utilized for designing the app's UI elements, ensuring an intuitive and accessible interface for visually impaired users.

6. Version Control System

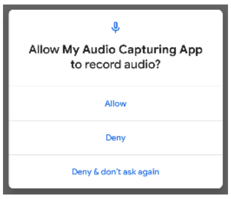
- **Git and GitHub:** Used for version control and collaboration. The source code and project files are hosted on GitHub to facilitate easy management, sharing, and updates.

GitHub Repository Link

GitHub Repository: https://github.com/Sumanth-Vallabhaneni-03/Voice-Navigation-app

5. PROPOSED MODEL

Voice-Navigation follows a well-structured model to facilitate seamless navigation, intuitive voice interaction, and accurate route guidance for visually impaired individuals. The model is designed to provide an accessible and user-friendly experience while ensuring reliability and security for all users.

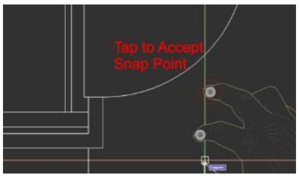


1. User Management & Authentication

- Secure application access through Android permissions to ensure microphone and internet access for voice processing.
- Permission management ensures that only authorized functionality (such as voice input and navigation) is accessible, enhancing user security.

2. Voice Input & Processing

- Users can input source and destination details using voice commands through the **SpeechRecognizer API**.
- The **RecognizerIntent** converts voice input to text, ensuring accurate location extraction.
- The system ensures that voice commands are processed efficiently, minimizing errors during navigation.



3. Tap-to-Select Start Point

- Users have the option to manually select their starting point by tapping anywhere on the screen.
- This feature allows flexibility in cases where the user prefers to choose a different starting point instead of the default detected location.

4. Audio Navigation System

- The app provides **step-by-step audio navigation cues** to guide users safely to their destination.
- The navigation system dynamically processes the route and delivers real-time voice instructions to ensure smooth and accurate guidance.

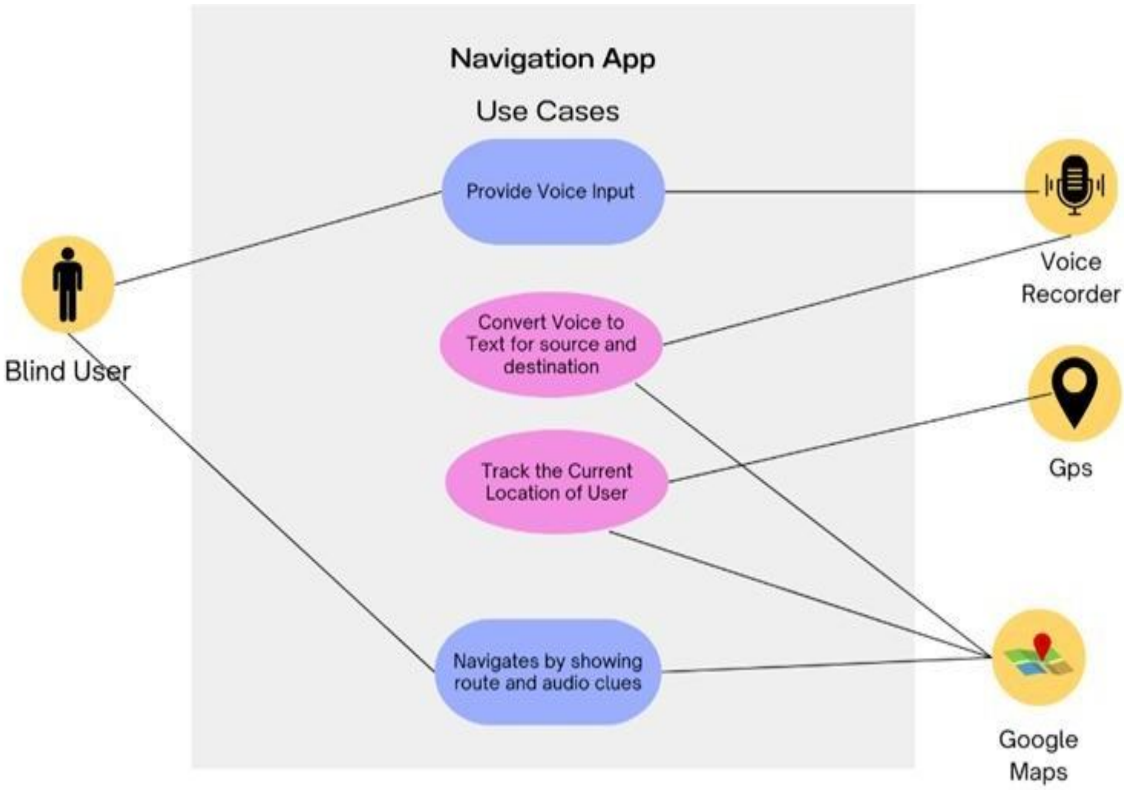
5. Google Maps Integration

- **Google Maps API** is used to generate optimal routes based on source and destination details provided by the user.
- The app seamlessly integrates with Google Maps, providing real-time route guidance and ensuring users reach their destination efficiently.

6. User-Friendly Interface

- The application features a clean, intuitive, and accessible interface optimized for visually impaired users.
- Simple navigation options ensure that users can effortlessly interact with the platform and initiate navigation using minimal input.

By incorporating these features, **Voice-Navigation** ensures a seamless, accessible, and secure navigation experience, empowering visually impaired individuals to explore their surroundings independently.



6. SAMPLE CODE

Folder structure

```
Voice-Navigation-app/
├── app/
├── build/
│   └── reports/
│       └── problems/
├── gradle/
├── .idea/
├── .gradle/
├── LICENSE
├── README.md
├── build.gradle.kts
├── gradle.properties
├── gradlew
├── gradlew.bat
├── local.properties
└── settings.gradle.kts
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
    <uses-permission android:name="android.permission.RECORD_AUDIO"/>
    <uses-permission android:name="android.permission.INTERNET"/>

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/icon"
        android:label="Voice Navigation"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.VoiceNavigationApp"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>
</manifest>
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
```

```xml
android:orientation="vertical"
android:background="#A68EF1">
<!--  #CCB5F8 light color  -->

<!-- App Name Header -->
<TextView
    android:id="@+id/appName"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Voice Navigation"
    android:textSize="24sp"
    android:textStyle="bold"
    android:textColor="@android:color/white"
    android:gravity="center"
    android:paddingTop="15dp"
    android:paddingBottom="10dp"/>

<!-- Full-Screen Clickable Area -->
<LinearLayout
    android:id="@+id/voiceButton"
    android:layout_width="match_parent"
    android:layout_height="0dp"
    android:layout_weight="1"
    android:gravity="center"
    android:orientation="vertical"
    android:clickable="true"
    android:focusable="true"
    android:background="@android:color/transparent">

    <TextView
        android:id="@+id/topText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Tap Anywhere to Start"
        android:textSize="24sp"
        android:textColor="@android:color/white"
        android:textStyle="bold"
```

```
            android:gravity="center"
            android:paddingBottom="50dp"/>
    </LinearLayout>

    <!-- Developer Name (Fixed at Bottom) -->
    <TextView
        android:id="@+id/developerName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Developed by: Sumanth"
        android:textSize="15sp"
        android:textColor="@android:color/white"
        android:gravity="center"
        android:paddingBottom="30dp"/>
</LinearLayout>
```

MainActivity.java

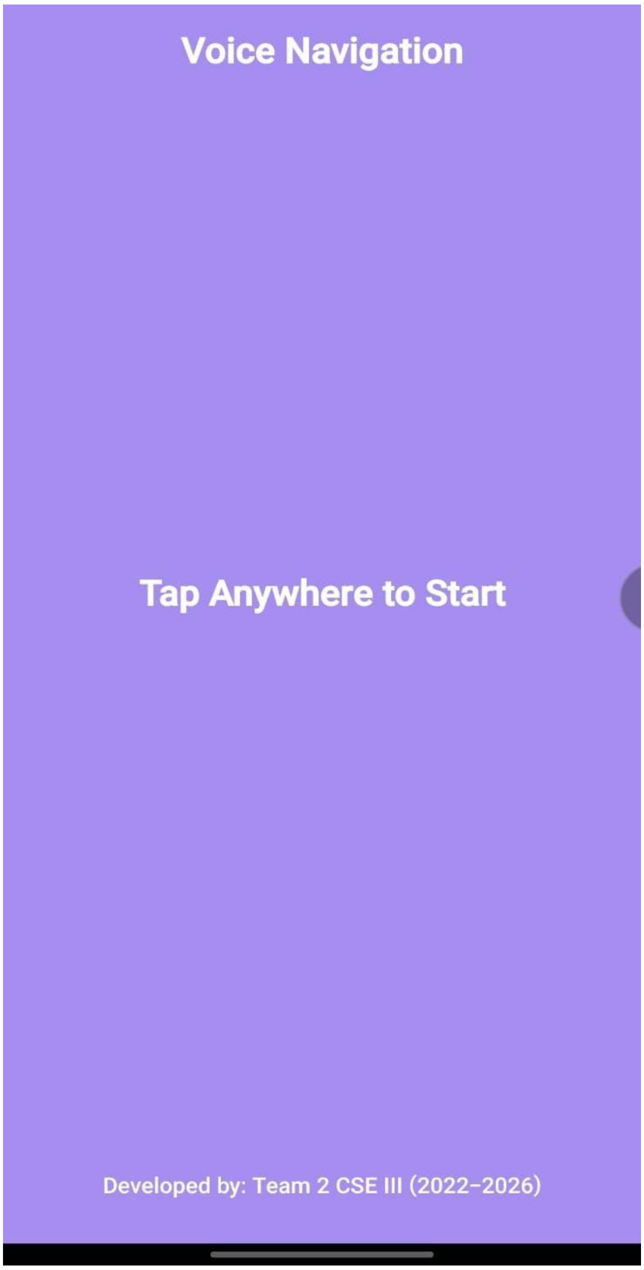
```
package com.example.voicenavigationapp;
import android.Manifest;
import android.content.Intent;
import android.content.pm.PackageManager;
import android.net.Uri;
import android.os.Bundle;
import android.speech.RecognizerIntent;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import java.util.ArrayList;
import androidx.annotation.NonNull;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;
import android.widget.LinearLayout;


public class MainActivity extends AppCompatActivity {
```

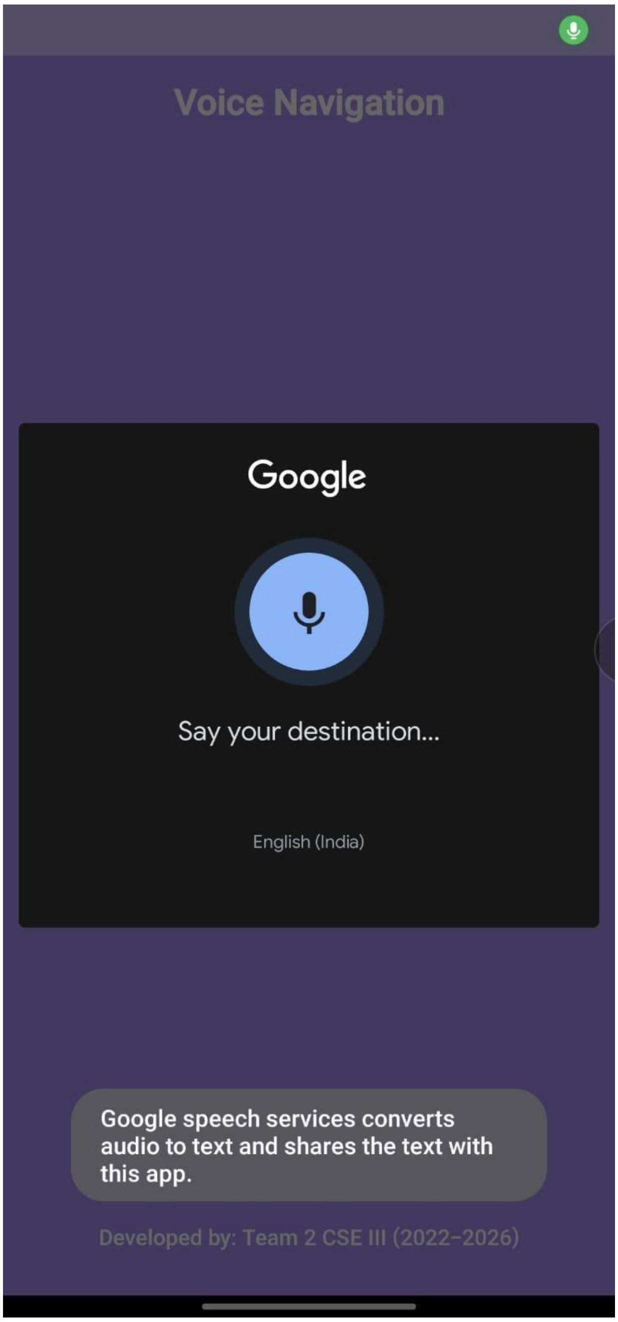
```
    private static final int SPEECH_REQUEST_CODE = 100;
    private static final int PERMISSION_REQUEST_CODE = 1;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        LinearLayout voiceButton = findViewById(R.id.voiceButton);

        // Request microphone permission
        if (ContextCompat.checkSelfPermission(this,
Manifest.permission.RECORD_AUDIO)
                != PackageManager.PERMISSION_GRANTED) {
            ActivityCompat.requestPermissions(this,
                    new String[]{Manifest.permission.RECORD_AUDIO},
PERMISSION_REQUEST_CODE);
        }

        // Make the entire screen clickable to start voice recognition
        voiceButton.setOnClickListener(v -> startListening());
    }

    private void startListening() {
        Intent intent = new
Intent(RecognizerIntent.ACTION_RECOGNIZE_SPEECH);

        // Set speech recognition model
        intent.putExtra(RecognizerIntent.EXTRA_LANGUAGE_MODEL,
RecognizerIntent.LANGUAGE_MODEL_FREE_FORM);
        intent.putExtra(RecognizerIntent.EXTRA_LANGUAGE, "en-IN"); // Indian
English
        intent.putExtra(RecognizerIntent.EXTRA_PARTIAL_RESULTS, true);

intent.putExtra(RecognizerIntent.EXTRA_SPEECH_INPUT_COMPLETE_SILE
NCE_LENGTH_MILLIS, 3000);
        intent.putExtra(RecognizerIntent.EXTRA_PROMPT, "Say your
```

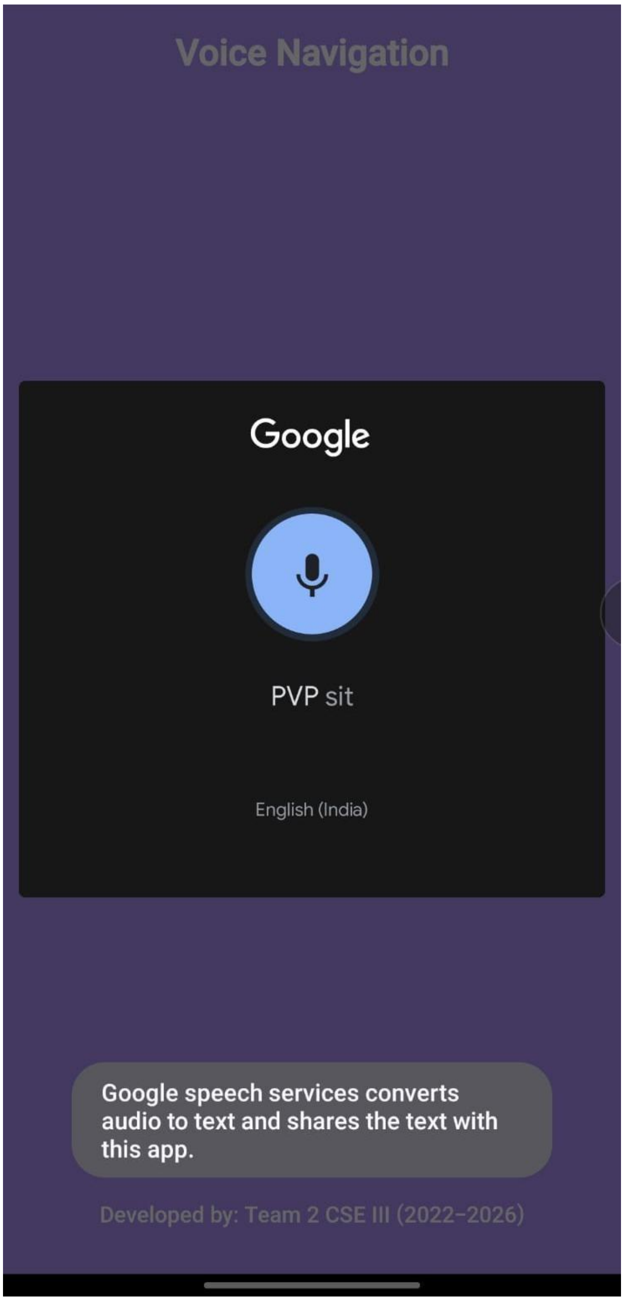
```
destination...");

        try {
            startActivityForResult(intent, SPEECH_REQUEST_CODE);
        } catch (Exception e) {
            Toast.makeText(this, "Speech recognition not supported on this device",
Toast.LENGTH_LONG).show();
        }
    }

    @Override
    protected void onActivityResult(int requestCode, int resultCode, Intent data)
{
        super.onActivityResult(requestCode, resultCode, data);

        if (requestCode == SPEECH_REQUEST_CODE && resultCode ==
RESULT_OK && data != null) {
            ArrayList<String> matches =
data.getStringArrayListExtra(RecognizerIntent.EXTRA_RESULTS);

            if (matches != null && matches.size() > 0) {
                String destination = matches.get(0);
                Toast.makeText(this, "You said: " + destination,
Toast.LENGTH_SHORT).show();
                openGoogleMaps(destination);
            }
        } else {
            Toast.makeText(this, "Speech recognition failed. Try again.",
Toast.LENGTH_SHORT).show();
        }
    }

    private void openGoogleMaps(String location) {
        Uri gmmIntentUri = Uri.parse("google.navigation:q=" +
Uri.encode(location));
        Intent mapIntent = new Intent(Intent.ACTION_VIEW, gmmIntentUri);
        mapIntent.setPackage("com.google.android.apps.maps");
```

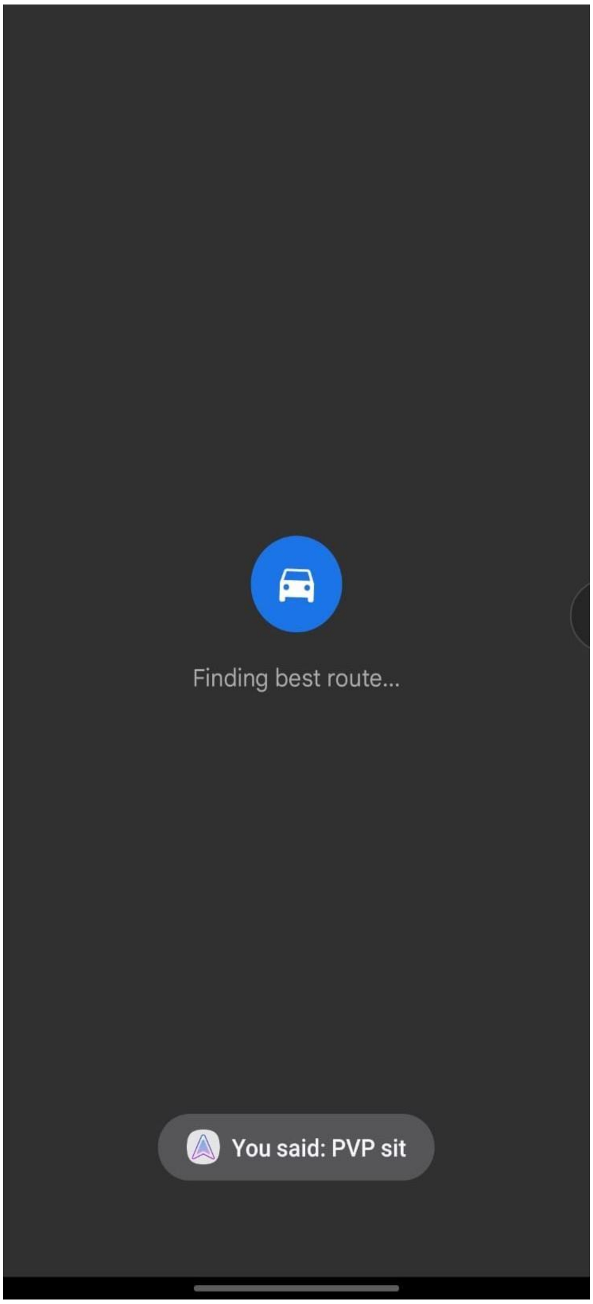
```
        startActivity(mapIntent);
    }

    // Handle permission request results
    @Override
    public void onRequestPermissionsResult(int requestCode, @NonNull
String[] permissions, @NonNull int[] grantResults) {
        super.onRequestPermissionsResult(requestCode, permissions,
grantResults);
        if (requestCode == PERMISSION_REQUEST_CODE) {
            if (grantResults.length > 0 && grantResults[0] ==
PackageManager.PERMISSION_GRANTED) {
                Toast.makeText(this, "Microphone Permission Granted",
Toast.LENGTH_SHORT).show();
            } else {
                Toast.makeText(this, "Microphone Permission Denied",
Toast.LENGTH_SHORT).show();
            }
        }
    }
}
```

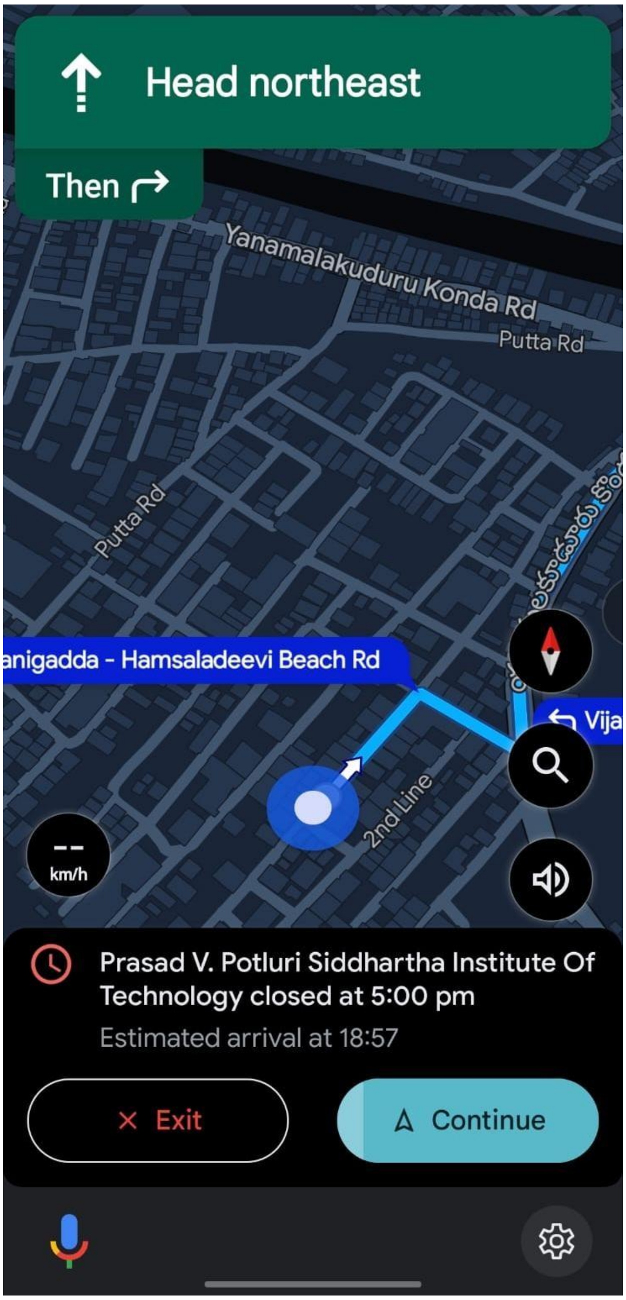
7. RESULT/OUTPUT SCREEN SHOTS

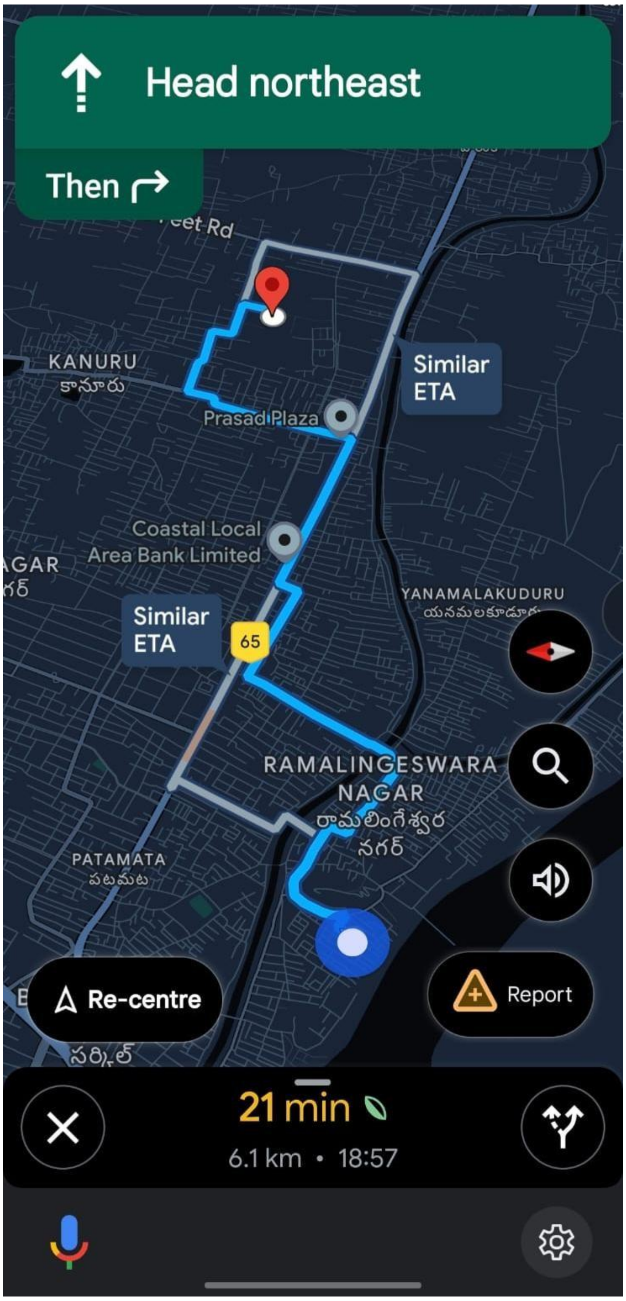





Voice Navigation
Google
PVP sit
English (India)
Google speech services converts audio to text and shares the text with this app.
Developed by: Team 2 CSE III (2022-2026)


Finding best route...
You said: PVP sit


Head northeast
Then
Yanamalakuduru Konda Rd
Putta Rd
Putta Rd
anigadda - Hamsaladeevi Beach Rd
Vija
2nd Line
--
km/h
Prasad V. Potluri Siddhartha Institute Of Technology closed at 5:00 pm
Estimated arrival at 18:57
Exit
Continue


Head northeast
Then
KANURU
Similar ETA
Prasad Plaza
Coastal Local Area Bank Limited
Similar ETA
65
YANAMALAKUDURU
RAMALINGESWARA NAGAR
PATAMATA
Re-centre
Report
21 min
6.1 km • 18:57

8. CONCLUSION AND FUTURE SCOPE

The Voice-Navigation application provides an accessible and user-friendly solution for visually impaired individuals to navigate their surroundings efficiently. By leveraging speech recognition technology and Google Maps integration, the app enables users to provide voice commands and receive step-by-step audio guidance. The system ensures ease of use by eliminating the need for manual input, making it an essential tool for individuals who rely on voice-based interaction.

This project demonstrates how modern technologies like speech recognition, real-time navigation, and intuitive UI design can significantly enhance accessibility. While the current implementation successfully assists users in voice-based navigation, future enhancements will further refine its functionality, offering a more comprehensive and intelligent navigation experience.

To expand the capabilities and improve the efficiency of **Voice-Navigation**, several enhancements can be implemented in future versions:

1. Offline Mode for Navigation

- Enable offline voice recognition and navigation for seamless operation without an internet connection.
- Store frequently used locations locally for quick access.

2. Smart Wearable Integration

- Integrate with smart devices like smartwatches or audio-based wearables to enhance accessibility.
- Provide haptic (vibration-based) feedback for better user awareness.

3. Enhanced Obstacle Detection

- Implement AI-based obstacle detection using external sensors like LiDAR or ultrasonic sensors.
- Provide real-time alerts about physical obstructions during navigation.

4. Public Transport Guidance

- Offer real-time updates on public transportation, including bus stops, train schedules, and metro routes.
- Guide users through multi-modal transportation systems with voice prompts.

8. REFERENCES

1. **Google Maps API Documentation** – Used for real-time navigation and route generation.

- URL: https://mapsplatform.google.com/

2. **SpeechRecognizer API Documentation** – Enables voice-to-text conversion for input processing.

- URL: https://developer.mozilla.org/en-US/docs/Web/API/SpeechRecognition

3. **Android Developer Documentation** – Reference for implementing Android permissions, intents, and UI interactions.

- URL: https://developer.android.com/develop

4. **RecognizerIntent Class Documentation** – Details about initiating and handling speech recognition.

- URL: https://developer.android.com/reference/android/speech/RecognizerIntent